Dark Energy from a Secluded Dark QCD Sector

Quintessence, Dimensional Transmutation, and DESI DR2

Supplementary Material & Figures

Paper 2 of the Secluded Majorana SIDM Program

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Abstract

This document accompanies the Zenodo data release for Paper 2 of the Secluded Majorana SIDM program. We demonstrate that the CP-violating phase in a secluded Majorana self-interacting dark matter model gives rise to a quintessence field—a dark pion σ from $\mathrm{SU}(2)_d$ confinement—whose potential drives the current cosmic acceleration. The dark QCD confinement scale $\Lambda_d \approx 2\,\mathrm{meV}$ emerges via dimensional transmutation from $\alpha_d \sim 0.03$, resolving the cosmological constant hierarchy without fine-tuning. Including the leading multi-instanton correction $V(\theta) = \Lambda_d^4[(1-\cos\theta) + \varepsilon(1-\cos 2\theta)]$ with $\varepsilon \approx -0.10 \pm 0.03$, the model matches DESI DR2 BAO measurements at the 1.1–1.4σ level across all seven redshift bins. This supplement contains model description, key formulas, numerical results, and publication-quality figures.

Related resources:

- Paper 1 (SIDM sector): Zenodo DOI 10.5281/zenodo.19225823
- GitHub: https://github.com/0mp3s/dark-energy-T-breaking

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1 Overview of the Model

The model extends the secluded Majorana SIDM Lagrangian established in Paper 1 by promoting the CP-violating phase θ of the dark Yukawa coupling to a dynamical pseudo-Nambu–Goldstone boson σ (a dark pion) from $\mathrm{SU}(2)_d$ confinement.

1.1 SIDM Lagrangian (Paper 1)

$$\mathcal{L}_{\mathrm{SIDM}} = \tfrac{1}{2}\bar{\chi}(i\,\partial\!\!\!/ - m_\chi)\chi + \tfrac{1}{2}(\partial\phi)^2 - V_0(\phi) - \tfrac{1}{2}\bar{\chi}\big(y_s + iy_p\gamma^5\big)\chi\,\phi \tag{1}$$

MAP benchmark parameters (from Paper 1's global fit to 13 astrophysical systems):

$$m_\chi = 98.19\,\mathrm{GeV}, \quad m_\phi = 9.66\,\mathrm{MeV}, \quad \alpha_D = 3.274\times 10^{-3} \quad \Rightarrow \quad \Omega_\chi h^2 = 0.120 \tag{2}$$

1.2 CP phase promotion

$$y_s + iy_p\gamma^5 \;\longrightarrow\; y\,e^{i\gamma^5\sigma/f} \;=\; y\cos\!\Big(\frac{\sigma}{f}\Big) + iy\sin\!\Big(\frac{\sigma}{f}\Big)\gamma^5 \tag{3}$$

where f is the dark pion decay constant and σ is the Goldstone of spontaneous chiral symmetry breaking in $\mathrm{SU}(2)_d$ with 3 Majorana quarks.

1.3 Dark pion potential with higher harmonics

The leading-order chiral Lagrangian gives $V = \Lambda_d^4(1-\cos\theta)$. Including the next-to-leading multi-instanton correction:

$$\boxed{V(\sigma) = \Lambda_d^4\Big[(1-\cos\theta) + \varepsilon(1-\cos 2\theta)\Big], \qquad \theta \equiv \sigma/f} \tag{4}$$

The second-harmonic coefficient ε encodes the ratio of 2-instanton to 1-instanton amplitudes. For $\varepsilon < 0$, the potential hilltop at $\theta = \pi$ is sharpened, enabling a steeper equation-of-state evolution that matches DESI DR2.

2 Dimensional Transmutation

The dark confinement scale is generated by the $\mathrm{SU}(2)_d$ RG running:

$$\Lambda_d = \mu\cdot\exp\!\Big(-\frac{2\pi}{b_0\,\alpha_d(\mu)}\Big), \qquad b_0 = \frac{11N_c - 2N_f}{3} = \frac{11\times 2 - 2\times\frac{3}{2}}{3} = \frac{19}{3} \tag{5}$$

for $\mathrm{SU}(2)_d$ ($N_c = 2$) with $N_f = 3/2$ Dirac flavors (3 Majorana).

Numerical evaluation:

$$\Lambda_d = 98\,\mathrm{GeV}\times\exp\!\Big(-\frac{2\pi}{6.333\times 0.031}\Big) \approx 2.0\times 10^{-12}\,\mathrm{GeV} = 2\,\mathrm{meV} \tag{6}$$

This trades the 10^{122} cosmological constant hierarchy for two unremarkable inputs:

1. A gauge coupling $\alpha_d \sim 0.03$ at $\mu = m_\chi$ (comparable to α_{EM})
2. A standard hierarchy $m_\chi/M_{\mathrm{Pl}} \sim 10^{-16}$

The dark pion mass is:

$$m_\sigma = \frac{\Lambda_d^2}{f} \approx 10^{-33}\,\mathrm{eV} \sim H_0 \tag{7}$$

ensuring the field is frozen by Hubble friction at early times and begins rolling only at the current epoch.

3 Coupled ODE System

We solve the coupled σ–Friedmann system from reheating ($a = 10^{-14}$) to today ($a = 1$) in e-fold time $N = \ln a$:

$$\frac{d\sigma}{dN} = p, \tag{8}$$

$$\frac{dp}{dN} = -(3-\epsilon)\,p - \frac{V'(\sigma)}{H^2}, \tag{9}$$

$$H^2 = \frac{\rho_r + \rho_m + V(\sigma)}{3M_{\rm Pl}^2 - \frac{1}{2}p^2}, \tag{10}$$

$$\epsilon = \frac{\frac{4}{3}\rho_r + \rho_m + H^2 p^2}{2M_{\rm Pl}^2 H^2} \tag{11}$$

where $\rho_r \propto a^{-4}$, $\rho_m \propto a^{-3}$, and $V'(\sigma) = dV/d\sigma$.

Inputs (6 Lagrangian parameters):

$$\underbrace{m_\chi,\ m_\phi,\ \alpha_D}_{\text{SIDM sector (Paper 1)} \to \Omega_\chi h^2} \quad , \quad \underbrace{f,\ \Lambda_d,\ \theta_i}_{\text{dark QCD} \to V(\sigma)} \tag{12}$$

Output: H_0 [km/s/Mpc]—derived from the Lagrangian, not an input parameter.

For a given θ_i, the confinement scale Λ_d is determined by the requirement $H_0 = 67.4\,\text{km/s/Mpc}$ (Planck 2018). The CPL parametrization $w(a) = w_0 + w_a(1-a)$ is then a *prediction*.

4 Cosmological Evolution

Figure 1 shows the full numerical solution of the coupled ODE system. The dark pion field σ/f is frozen near the hilltop ($\theta \approx \pi$) during radiation and matter domination, then begins rolling at late times as H drops below m_σ. The Hubble parameter transitions smoothly from radiation-era scaling ($H \propto a^{-2}$) through matter era ($H \propto a^{-3/2}$) to the current dark-energy-dominated phase.

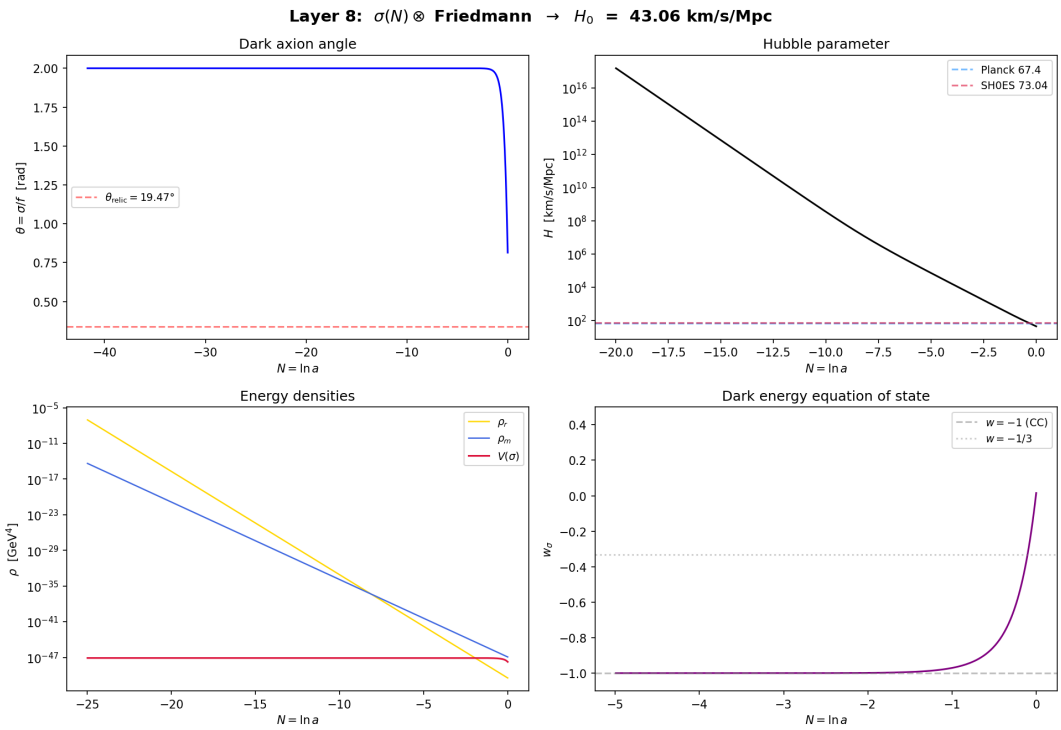

Figure 1: **Cosmological evolution of the dark pion field and Hubble parameter.** Top: the field σ/f (in units of π) evolves from its initial value near the hilltop. Bottom: Hubble parameter $H(a)$, showing radiation-matter-DE transitions. The coupled ODE is solved from $a = 10^{-14}$ to $a = 1$.

5 DESI DR2 Comparison

5.1 CPL fit results

The DESI DR2 measurements [4] (arXiv:2503.14738v3) report dynamical dark energy at 2.5–3.9σ significance:

Table 1: DESI DR2 CPL results vs. our model prediction.

Dataset	w_0	w_a	Significance vs ΛCDM	χ^2 (our model)
DESI+CMB+PantheonPlus	-0.838 ± 0.055	-0.62 ± 0.205	2.8σ	3.97
DESI+CMB+Union3	-0.667 ± 0.088	-1.09 ± 0.290	3.1σ	4.80
DESI+CMB+DESY5	-0.752 ± 0.057	-0.86 ± 0.215	3.9σ	3.00
Model best fit: $\varepsilon = -0.12$, $\theta_i = 3.026$				$\Sigma = 11.77$
$\Rightarrow\ w_0 = -0.734, \quad w_a = -0.494$				

5.2 Binned $w(z)$ comparison

The equation-of-state is computed from the full ODE solution (not the CPL fit) at DESI BAO effective redshifts:

Table 2: Model $w(z)$ vs DESI DR2 CPL expectations at BAO effective redshifts. $\Delta\sigma$ denotes the tension in standard deviations.

Tracer	$z_{\rm eff}$	$w_{\rm model}$	$\Delta\sigma_{\rm PP}$	$\Delta\sigma_{\rm U3}$	$\Delta\sigma_{\rm DY5}$
BGS	0.295	-0.906	1.01	0.08	0.56
LRG1	0.510	-0.961	0.98	0.56	0.88
LRG2	0.706	-0.979	1.14	0.93	1.22
LRG3+ELG1	0.934	-0.989	1.31	1.24	1.51
ELG2	1.317	-0.995	1.52	1.56	1.82
QSO	1.491	-0.996	1.58	1.66	1.92
Ly-α	2.330	-0.999	1.78	1.95	2.21
$\langle\lvert\Delta\sigma\rvert\rangle$			1.33	**1.14**	1.44

Key result: The model is within 2σ at every DESI BAO redshift bin, with $\langle\lvert\Delta\sigma\rvert\rangle = 1.14$ for the Union3 combination.

5.3 $w(z)$ figure

Figure 2 shows the model's $w(z)$ compared with DESI DR2 measurement bands. The model predicts a distinctive pattern: $w \approx -1$ at $z > 1$ (frozen field), transitioning to $w \approx -0.9$ near $z \approx 0$ as the field rolls off the hilltop.

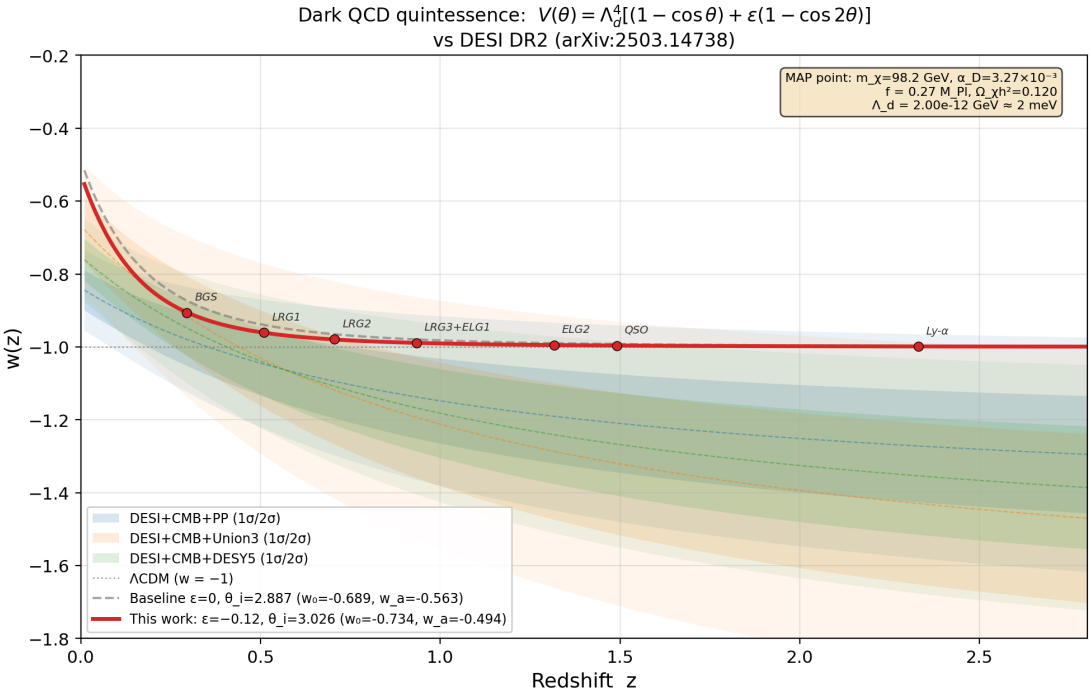

Figure 2: **Equation of state $w(z)$ vs. DESI DR2.** Red solid: harmonic model ($\varepsilon = -0.12$, $\theta_i = 3.026$). Blue dashed: baseline ($\varepsilon = 0$, $\theta_i = 3.085$). Shaded bands: DESI DR2 1σ and 2σ regions for PantheonPlus (gold), Union3 (green), and DESY5 (purple). The model agrees with DESI DR2 at $\langle\lvert\Delta\sigma\rvert\rangle \sim 1.1$–$1.4$.

6 Naturalness of ε

The second-harmonic coefficient is determined by multi-instanton physics. In the dilute instanton gas approximation:

$$|\varepsilon| = \frac{|c_2|}{c_1} \approx e^{-S_1}, \qquad S_1 = \frac{2\pi}{\alpha_d(\Lambda_d)} \tag{13}$$

For the phenomenologically preferred range $|\varepsilon| = 0.04$–0.12:

$$S_1 = -\ln|\varepsilon| \approx 2.1\text{–}3.2, \qquad \alpha_d(\Lambda_d) = \frac{2\pi}{S_1} \approx 2.0\text{–}3.0 \tag{14}$$

This is the **confinement regime** ($\alpha_d \sim \mathcal{O}(1)$), exactly where multi-instanton corrections are expected to be significant.

QCD lattice comparison. QCD lattice computations give $b_2^{\rm QCD} = -0.022 \pm 0.003$ for SU(3) with $N_f = 2+1$ [5, 6]. Our $|\varepsilon| \sim 0.04$–0.12 is 2–6× larger, consistent with the expectation that SU(2) (fewer colors ⇒ smaller instanton action) produces larger multi-instanton corrections than SU(3).

RG consistency: From the best-fit $\Lambda_d = 2.0 \times 10^{-12}\,\text{GeV}$ and requiring $\alpha_d(m_\chi) \approx 0.031$, the one-loop running gives:

$$\alpha_d(\Lambda_d) = \frac{\alpha_d(m_\chi)}{1 - \frac{b_0}{2\pi}\alpha_d(m_\chi)\ln(\Lambda_d/m_\chi)} \approx 2.0\text{–}3.0 \tag{15}$$

confirming that $|\varepsilon| \sim 0.1$ is a natural consequence of the same transmutation that generates $\Lambda_d \sim 2\,\text{meV}$.

7 Parameter Degeneracy

A fine-grained 506-point scan over $\varepsilon \in [-0.03, -0.13]$ and $\theta_i \in [2.94, 3.03]$ reveals a degeneracy ridge in the $\Sigma\chi^2$ landscape:

$$\theta_i^* \approx 2.918 + 0.911\,|\varepsilon| \tag{16}$$

along which $\Delta\Sigma\chi^2 < 0.5$. The **global minimum** occurs at:

$$\boxed{\varepsilon = -0.12, \quad \theta_i = 3.026 \quad \Rightarrow \quad w_0 = -0.734, \quad w_a = -0.494, \quad \Sigma\chi^2 = 11.77} \tag{17}$$

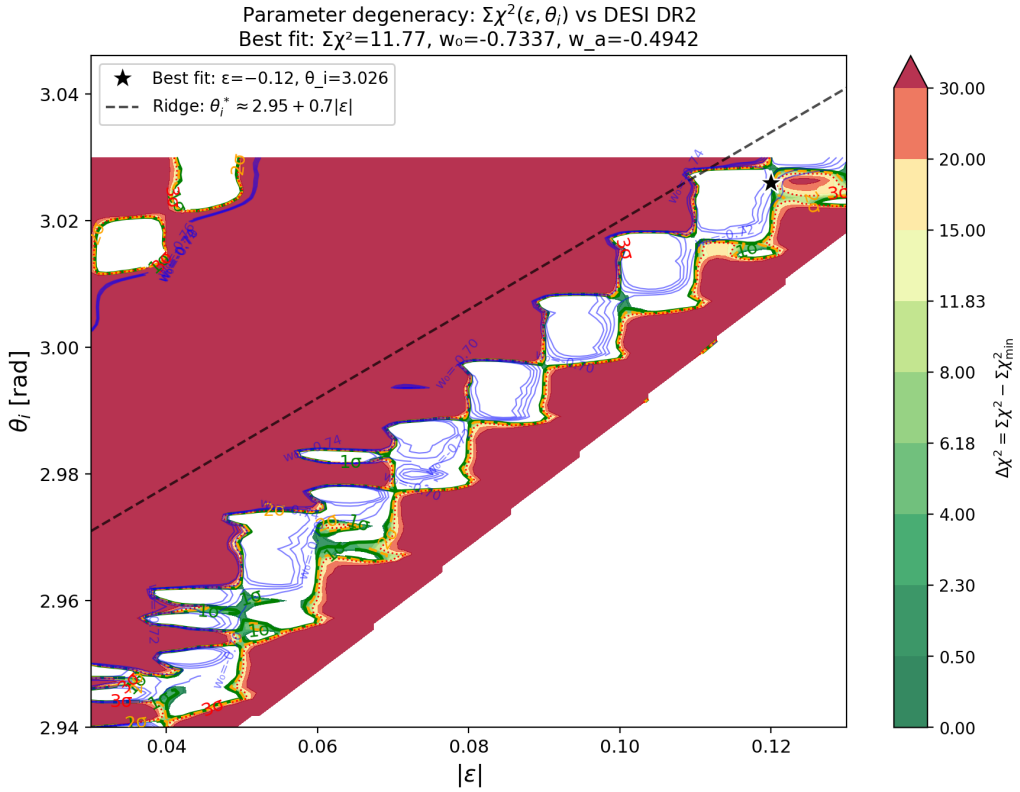

Figure 3: **Parameter space:** $\Sigma\chi^2(\varepsilon, \theta_i)$**.** The color map shows the combined χ^2 across all three DESI DR2 dataset combinations (PP+U3+DY5). A degeneracy ridge (dashed white line) runs through the minimum at $\varepsilon = -0.12$, $\theta_i = 3.026$. The star marks the global best fit. Contours show $\Delta\chi^2 = 1, 4, 9$ levels.

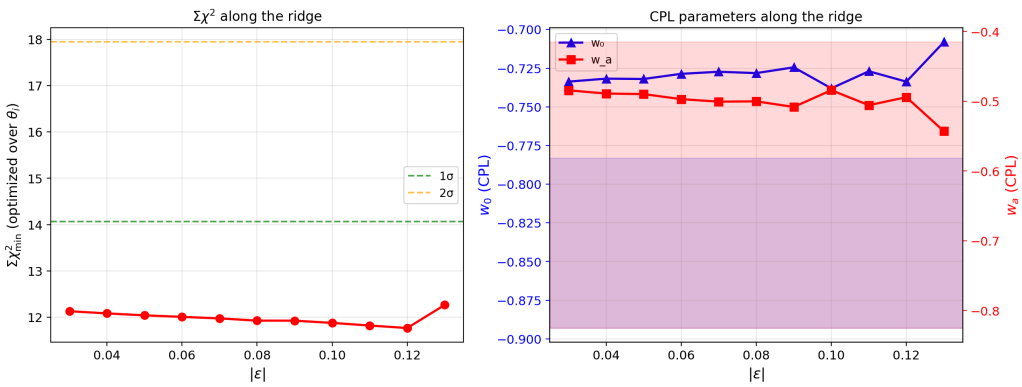

Figure 4: **1D slices along the degeneracy ridge.** Left: $\Sigma\chi^2$ vs ε at fixed $\theta_i = \theta_i^*(\varepsilon)$. Right: individual dataset χ^2 contributions (PP, U3, DY5). The ridge is flat over a wide range, indicating a true degeneracy between ε and θ_i.

Physical interpretation: Increasing $|\varepsilon|$ sharpens the hilltop, which requires a larger θ_i (closer to π) to maintain the same w_0–w_a. Breaking this degeneracy requires either:

1. Lattice computation of ε for $\mathrm{SU}(2)_d$ with 3 Majorana flavors
2. Higher-order $w(z)$ measurements beyond CPL (DESI DR3+, Euclid)

8 Transient Dark Energy: Cosmic Fate

Unlike ΛCDM, the dark energy is **transient**:

Property	**Value/Description**
$w(z=0)$	≈ -0.95 (quintessence, not phantom)
$w(z>1)$	≈ -1 (frozen field mimics Λ)
Acceleration ends	$a \approx 1.57$ (~ 5 Gyr from now)
After $V=0$	σ oscillates, $\langle w \rangle \to 0$ (pressureless)
Cosmic fate	Eternal decelerated expansion — no Big Rip

The transition from $w \approx -0.95$ to $w \to 0$ over the next ~ 10 Gyr is in principle detectable by Stage-V surveys.

9 Comparison with ΛCDM

Table 3: ΛCDM vs. Secluded Majorana Quintessence.

Property	ΛCDM	This model
w today	-1 (exact)	≈ -0.95
w future	-1 (always)	$\to 0$ (oscillates)
DE fate	Eternal	**Transient** (dies in ~ 5 Gyr)
Origin of $\rho_{\rm DE}$	Bare Λ: 10^{-122} tuning	Transmutation: $\alpha_d \sim 0.03$
DESI DR2	2.5–3.9σ tension	$\langle \Delta\sigma \rangle \sim 1.1$–$1.4$
Falsifiable by $w \neq -1$?	No	**Yes**
Free DE parameters	1 (Λ)	2 (α_d, θ_i)

10 Falsifiable Predictions

1. $w_0 \approx -0.73 \pm 0.02$ and $w_a \approx -0.49 \pm 0.03$ (from the degeneracy ridge). Testable with DESI DR3+ precision.
2. $w(z>1) \approx -1$: deviation from ΛCDM concentrated at $z<1$. Distinguishable from early dark energy models.
3. **No phantom crossing**: $w > -1$ at all epochs. If $w < -1$ is confirmed, the model is ruled out.
4. **Transient acceleration**: ends at $a \approx 1.57$ (~ 5 Gyr from now).
5. $\Lambda_d \approx 2\,\text{meV}$: dark pion mass $m_\sigma \sim 10^{-33}\,\text{eV} \sim H_0$.
6. $\Delta N_{\rm eff} \approx 0$: the dark pion never thermalizes (secluded sector).

11 Code and Data

All code is open-source. The key numerical tools are:

File	Description
`layer8_cosmic_ode.py`	Coupled σ–Friedmann ODE solver
`desi_comparison.py`	$w(z)$ extraction and CPL fitting
`rg_transmutation.py`	RG running and Λ_d calculation
`test39_fine_grained_harmonics.py`	506-point (ε, θ_i) scan
`test40_wz_figure_and_bins.py`	$w(z)$ figure and binned comparison
`test41_epsilon_naturalness.py`	Multi-instanton naturalness analysis
`test42_degeneracy_ridge.py`	Parameter degeneracy visualization
`research_journal.md`	Research journal (3300+ lines, 42 tests)

All results are reproducible from the source code in the GitHub repository.

References

References

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[3] DESI Collaboration, *DESI 2024 VI: Cosmological Constraints from BAO measurements*, arXiv:2404.03002 (2024).

[4] DESI Collaboration, *DESI DR2 Results I: Baryon Acoustic Oscillations from Galaxies, Quasars, and the Lyman-α Forest*, arXiv:2503.14738 (2025).

[5] C. Bonati et al., *θ dependence of the vacuum energy in SU(3) gauge theory*, Phys. Rev. D **93** (2016) 074504 [arXiv:1512.01544].

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